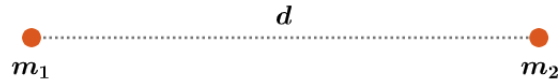


2007 F=ma Exam: Problem 8

Kevin S. Huang



Recall the gravitational potential energy between two masses is given by

$$U = -\frac{Gm_1m_2}{r}$$

under the standard convention that $U \rightarrow 0$ as $r \rightarrow \infty$. We have

$$U = -\frac{Gm_1m_2}{d}$$

Thus,

$$U' = -\frac{Gm_1m_2}{2d} = \frac{U}{2}$$

so the answer is B.